

Artificial Intelligence

AI-2002

Lecture 5

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Lecture Department of Computer Science

FAST NUCES CFD

Informed Searches

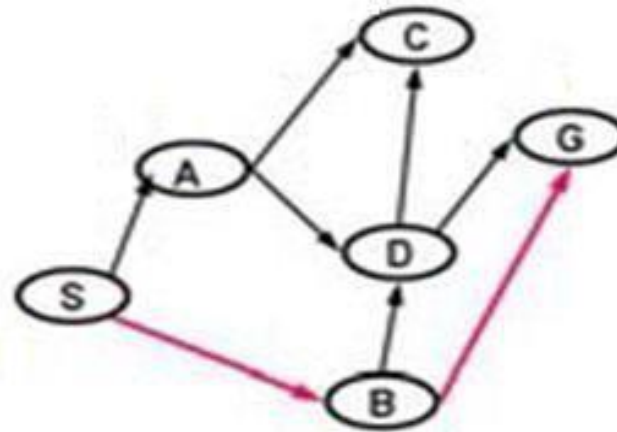
- One that uses **problem-specific knowledge** beyond the definition of the problem itself.
- For Example
 - Best First Search
 - A* Search

Best First Search

- A node is selected for expansion based on an evaluation function, $f(n)$.
 - the *lowest* evaluation is expanded first
- Best-first search algorithms include as a component of
- f a **heuristic function**, denoted $h(n)$.
- $h(n)$ = estimated cost of the cheapest path
 - $h(n)$ takes a *node* as input, but, unlike $g(n)$, it depends only on the *state* at that node.

$$f(n) = h(n)$$

Best First Search

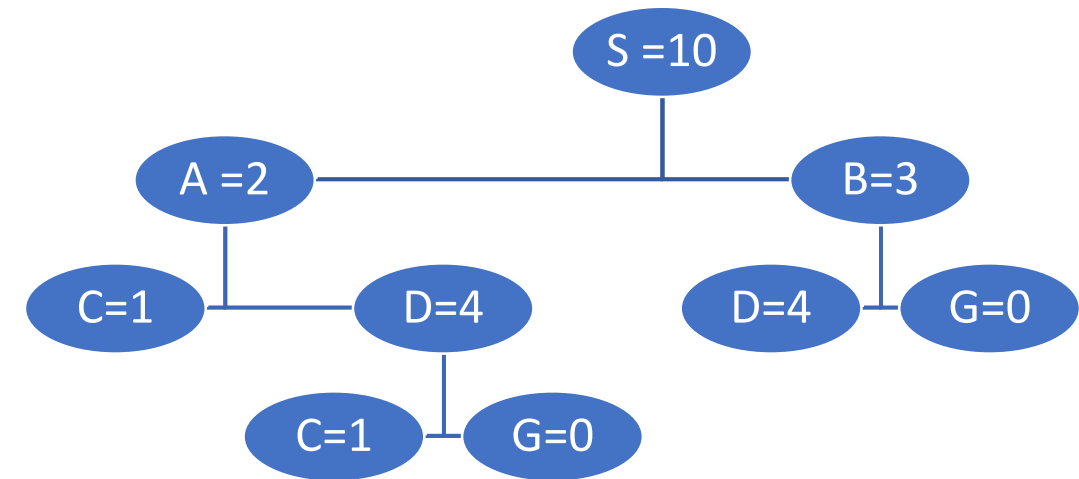


Heuristic Values

A=2	C=1	S=10
B=3	D=4	G=0

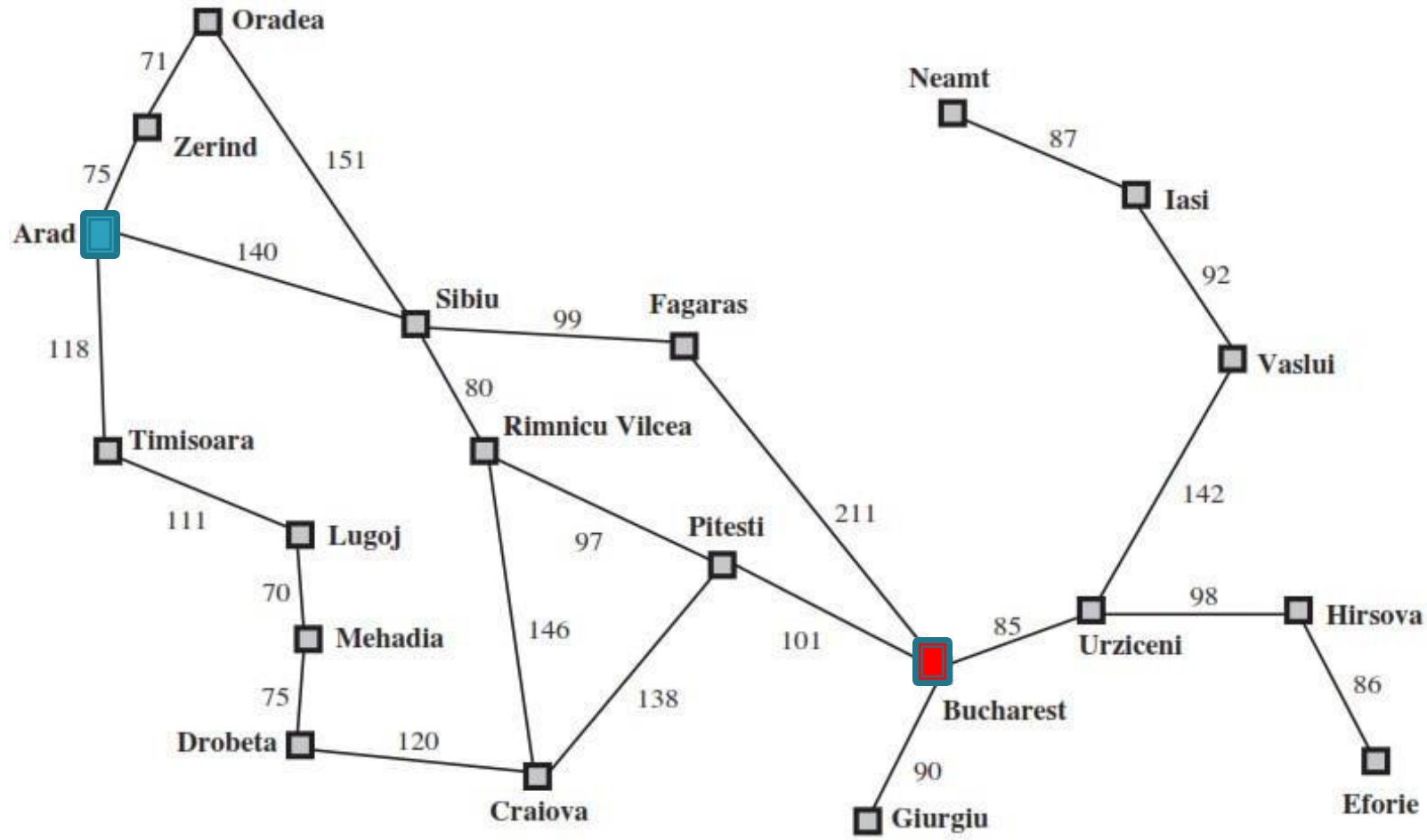
Implementation of Best First Search

	Q	Visited
1	(10 S)	S
2	(2 A S) (3 B S)	A, B, S
3	(1 C A S) (3 B S) (4 D A S)	C, D, B, A, S
4	(3 B S) (4 D A S)	C, D, B, A, S
5	(0 G B S) (4 D A S)	G, C, D, B, A, S



4DBS is not included in the list because we have already visited the D from 4DAS.

Example:



Asimplified road map of part of Romania.

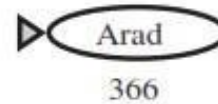
Values of Heuristics

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

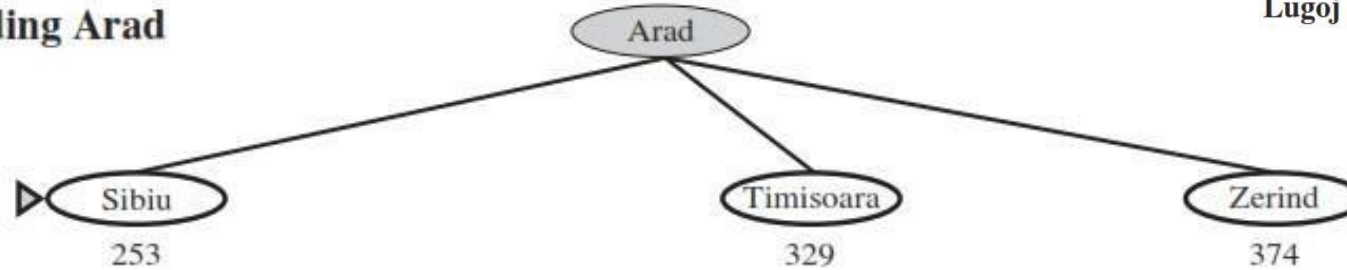
Best First Search

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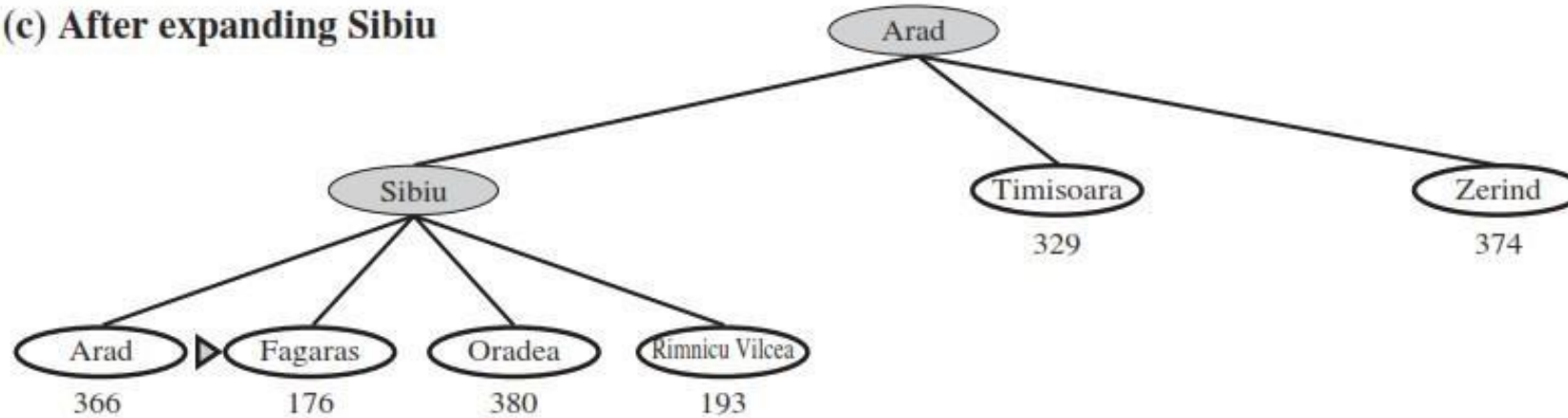
(a) The initial state



(b) After expanding Arad

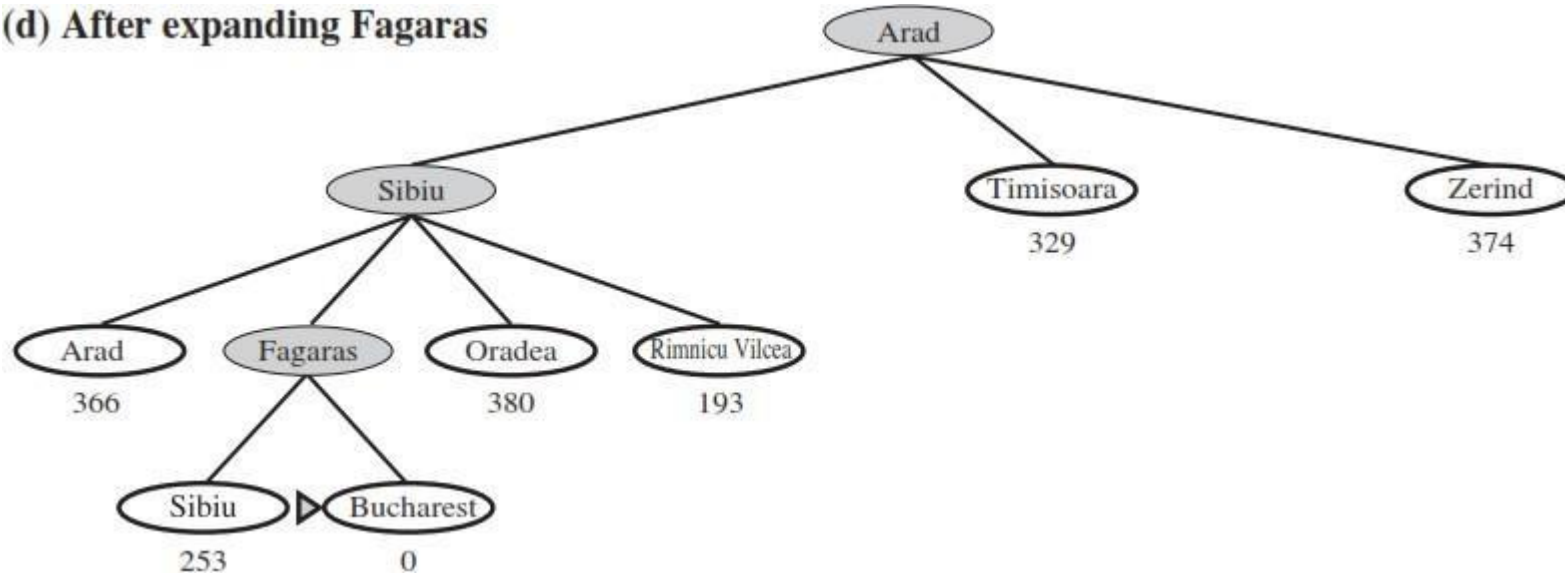


(c) After expanding Sibiu



Best First Search

(d) After expanding Fagaras



Stages in a greedy best-first tree search for Bucharest with the straight-line distance heuristic. Nodes are labeled with their h-values.

Best First Search

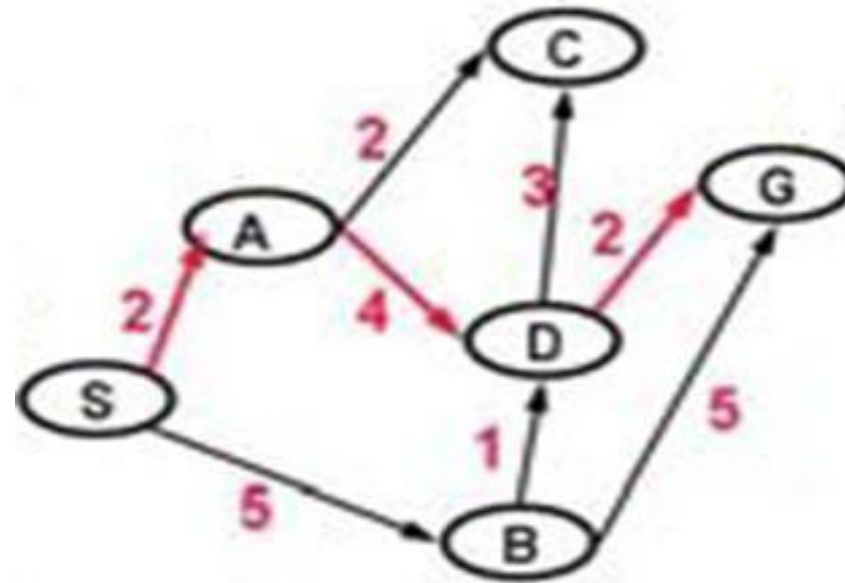
- Time and Space Complexity
- The *worst-case time and space* complexity for the tree version is $O(b^m)$, where m is the maximum depth of the search space.
 - With a good heuristic function, however, the complexity can be reduced substantially

A* Search

- We **can bias Uniform-cost search** to find the shortest path to the goal.
- In fact, we are interested in by using a **heuristic function $h(n)$** which is an estimate of the distance from a state to the goal.
- It evaluates nodes by **combining $g(n)$** , the cost to reach the node, and **$h(n)$** , the cost to get from the node to the goal:

$$f(n) = g(n) + h(n)$$

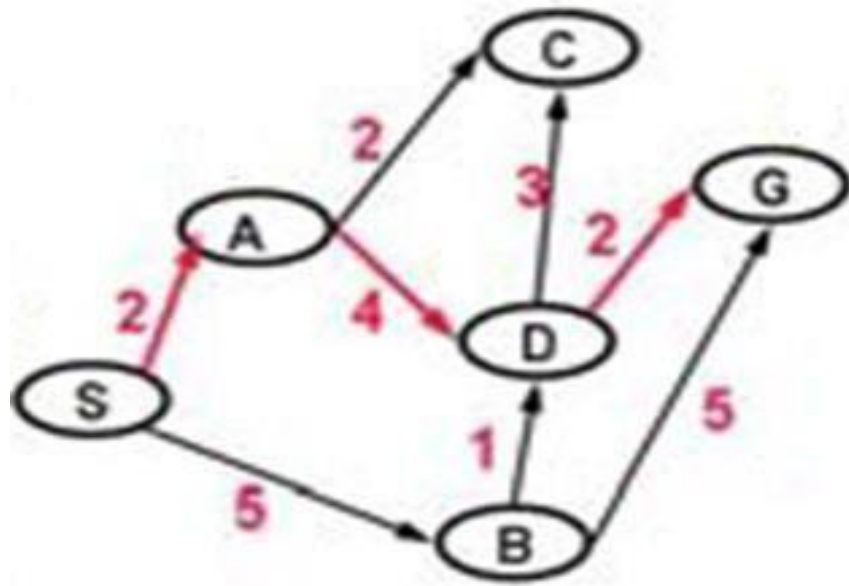
A* Search



Heuristic Values

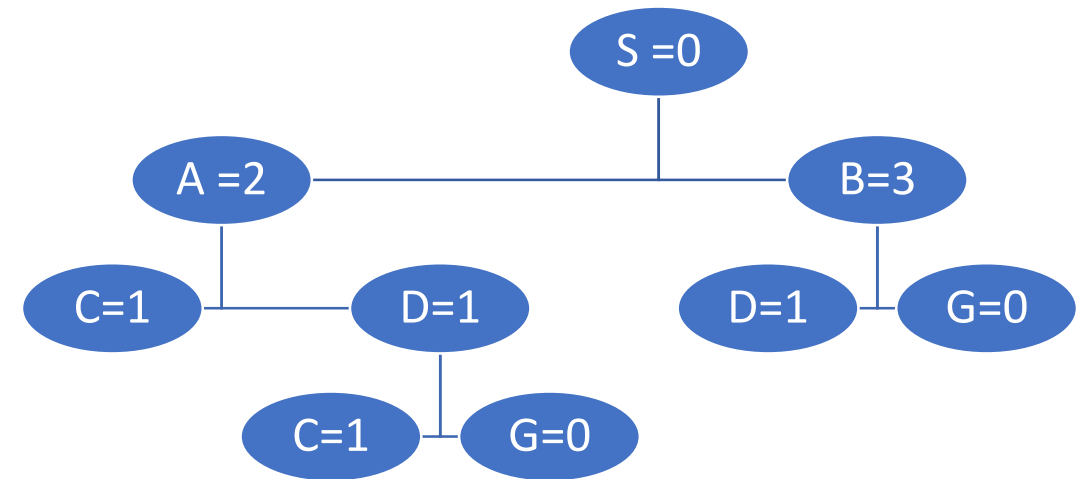
A=2	C=1	S=0
B=3	D=1	G=0

A* Search



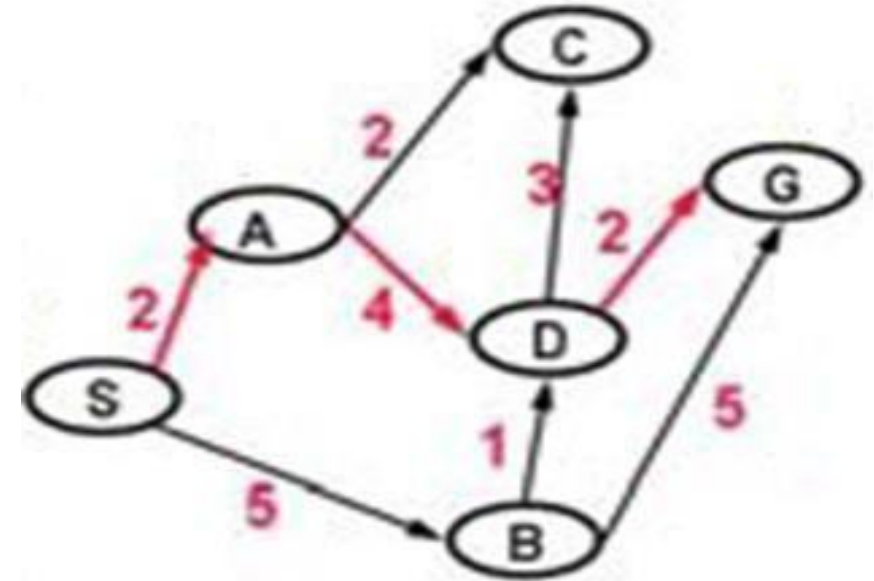
Heuristic Values

A=2	C=1	S=0
B=3	D=1	G=0



A* Search

	Q
1	(<u>0 S</u>)
2	(<u>4 A S</u>) (8 B S)
3	(<u>5 C A S</u>) (7 D A S) (8 B S)
4	(<u>7 D A S</u>) (8 B S)
5	(<u>8 G D A S</u>) (10 C D A S) (8 B S)



Heuristic Values

A=2

C=1

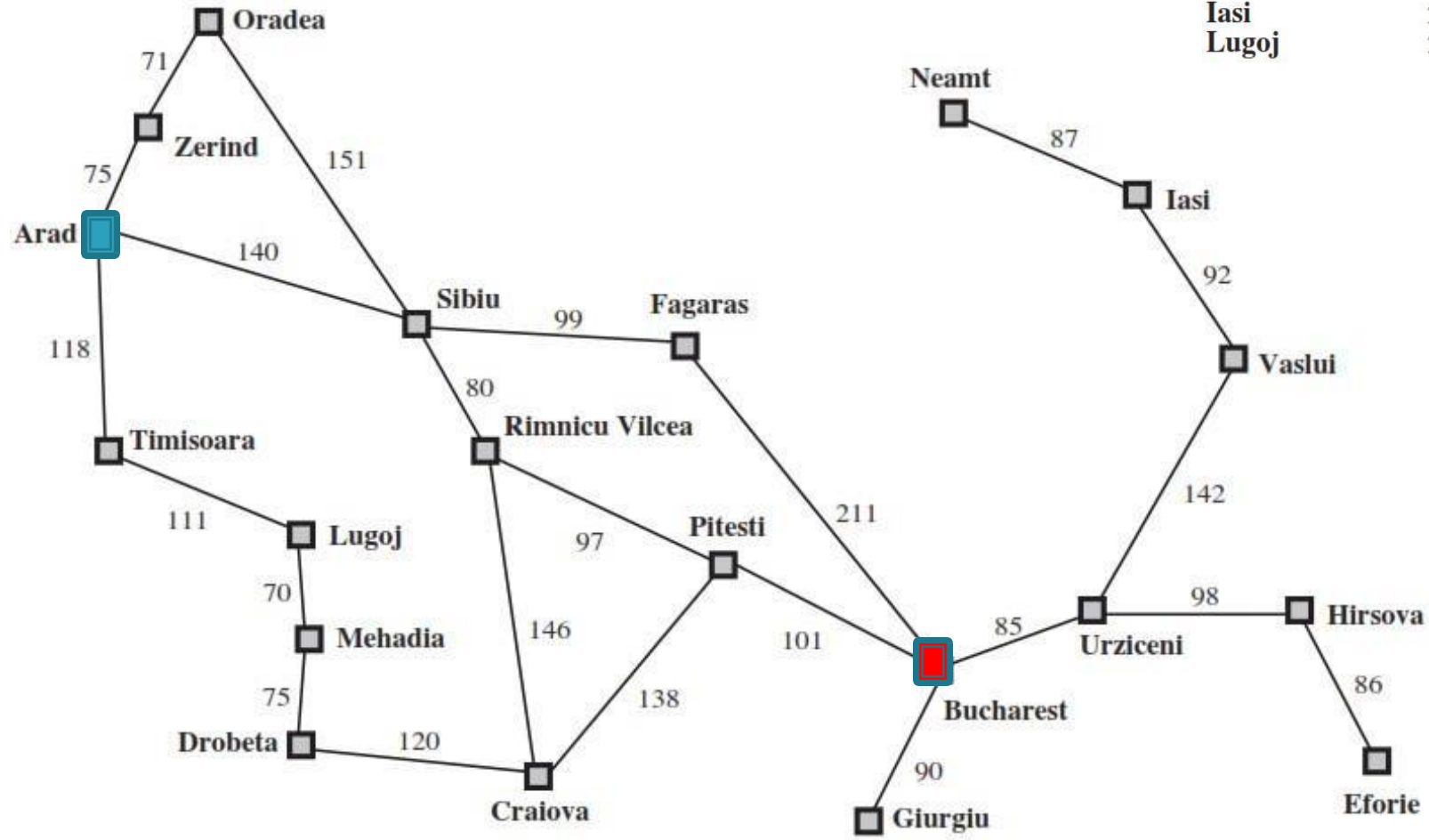
S=0

B=3

D=1

G=0

A* Search



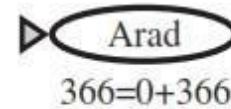
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Heuristics Values

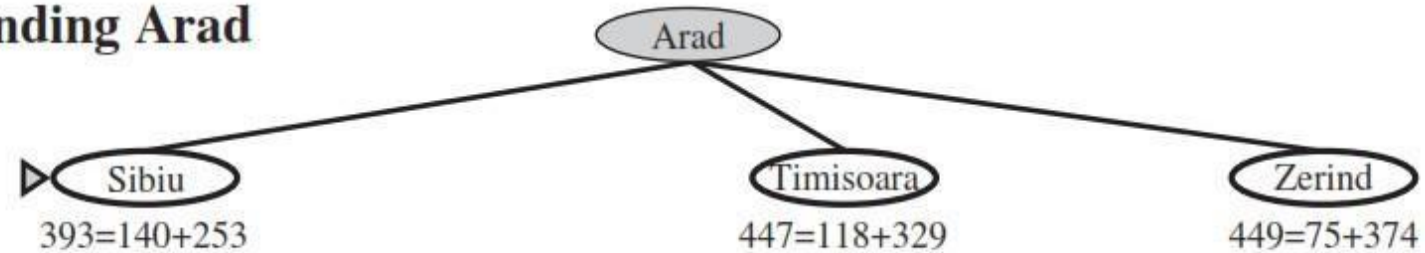
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Example

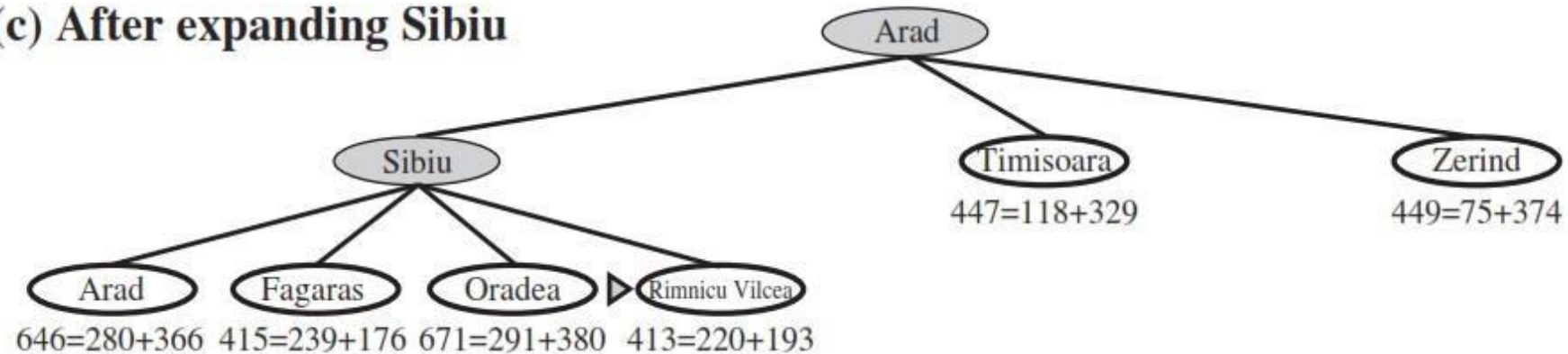
(a) The initial state



(b) After expanding Arad

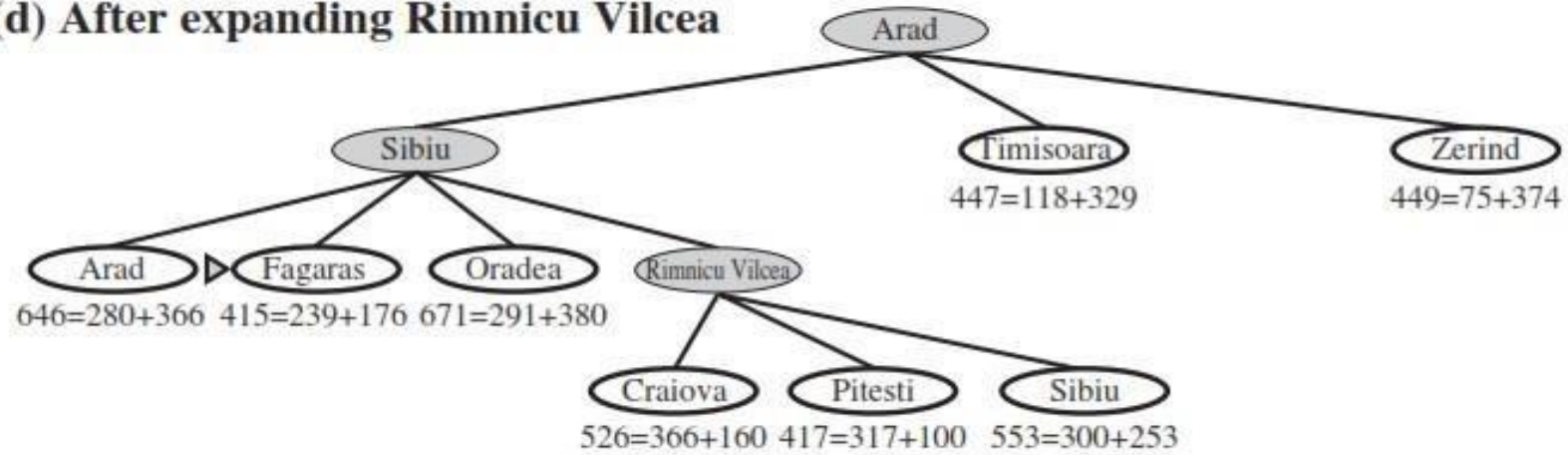


(c) After expanding Sibiu

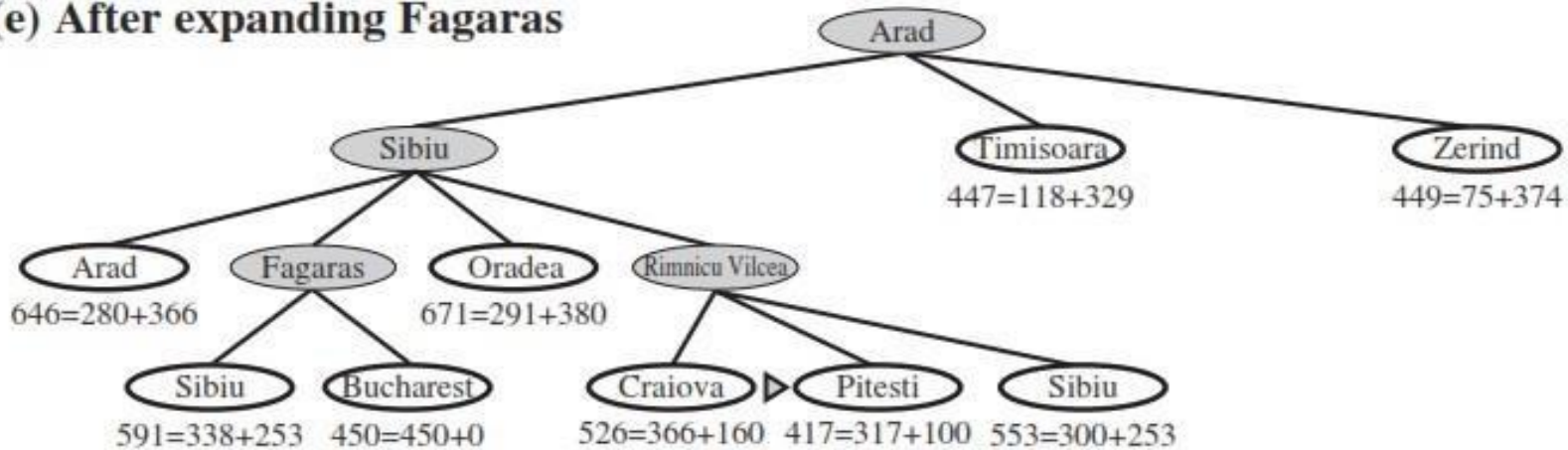


Example:

(d) After expanding Rimnicu Vilcea

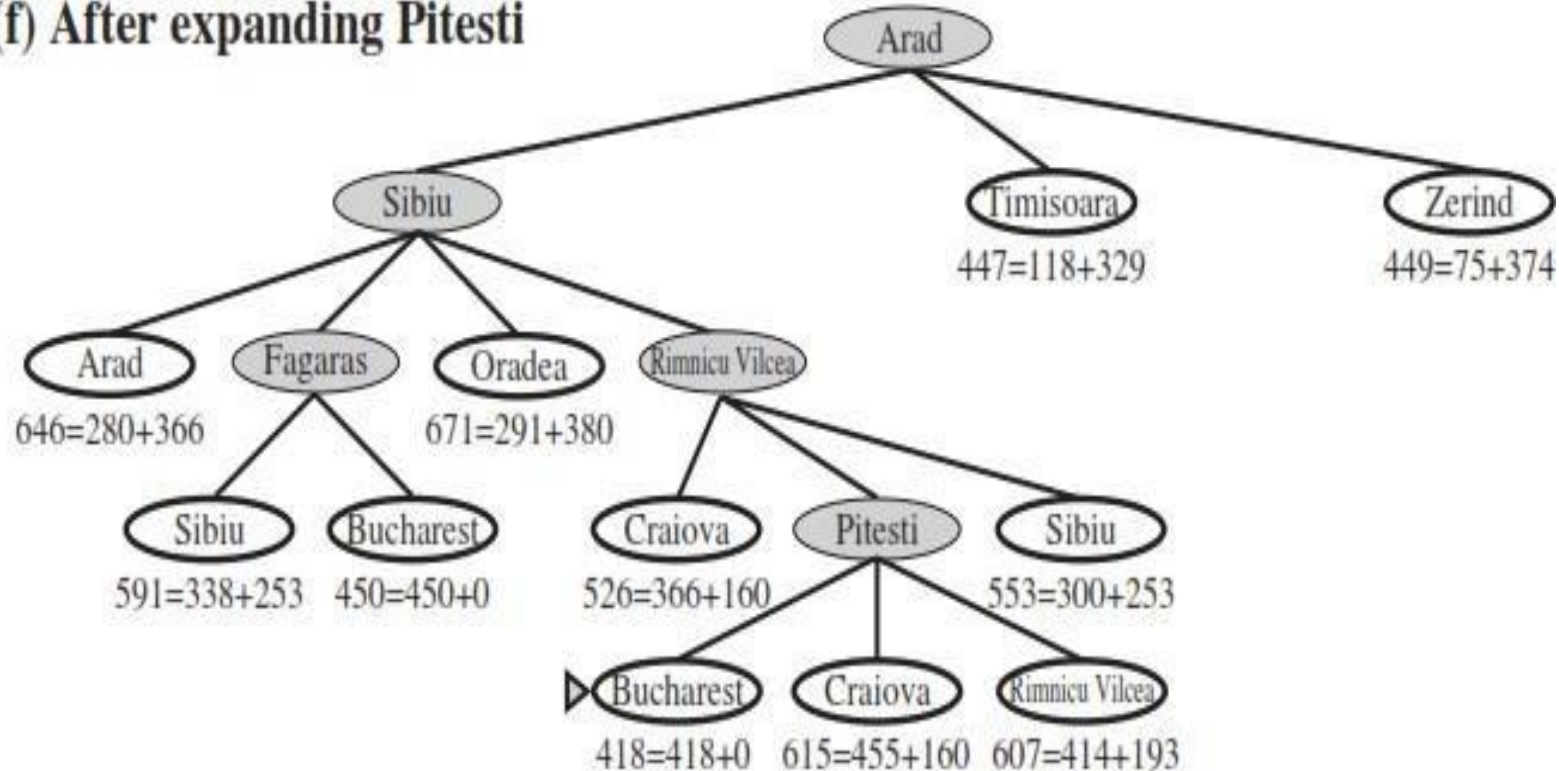


(e) After expanding Fagaras



Example:

(f) After expanding Pitesti



**A* search for Bucharest: Nodes are labeled with $f = g + h$.
The h values are the straight-line distances to Bucharest.**

A* Search

- In fact, we are interested in by using a **heuristic function $h(n)$** which is an estimate of the distance from a state to the goal.
- It evaluates nodes by combining **$g(n)$** , the cost to reach the node, and **$h(n)$** , the cost to get from the node to the goal:

$$f(n) = g(n) + h(n)$$

A* Prosperities

- Admissibility
- Consistency

Heuristics Admissibility

Straight line distance is an admissible estimate for path length

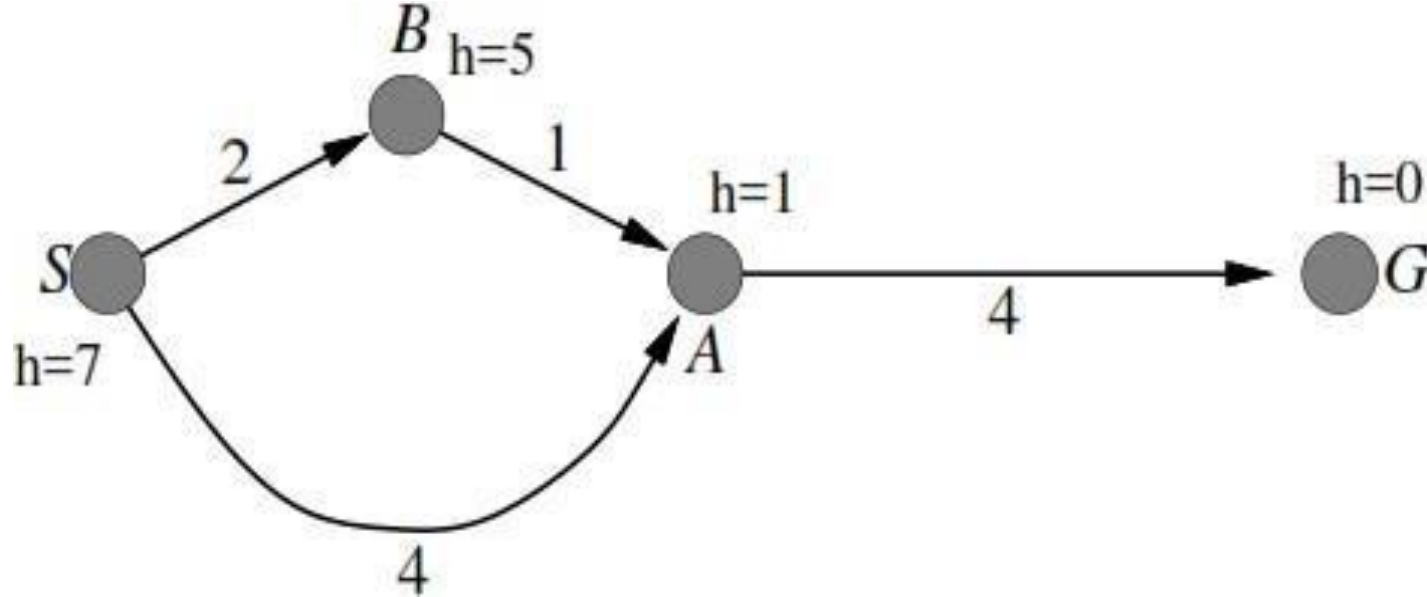
- in euclidian space.
- • Uniform cost search is an instance of A^* , If we set
- $h(n) = 0$.
 - Use of an admissible estimate guarantee that Uniform-cost search will find the shortest path.

Uniform-cost search with admissible estimate (heuristic) is known as A^* search.

$$h'(n) \leq h(n)$$

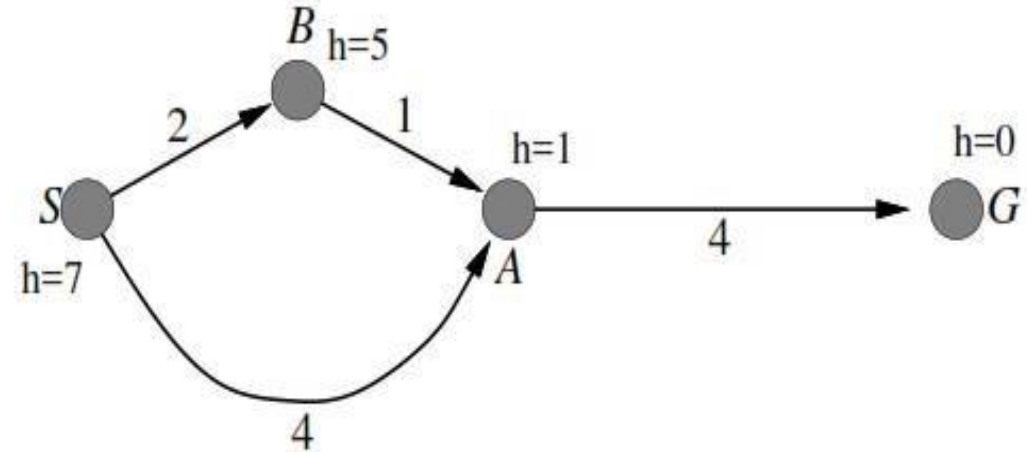
Example

Are all heuristics admissible and consistent in this graph?



Example

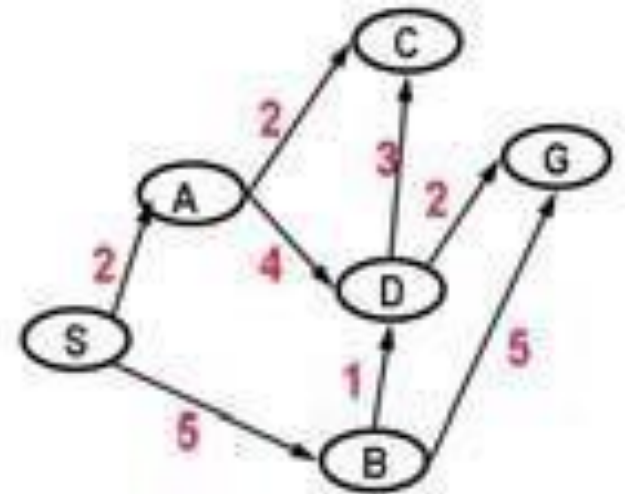
- An estimate that always **underestimates the real path length to the goal** is called **admissible (heuristic) estimate**.
- **Admissibility:**
 - ❑ (S): $7 \leq 7, 7 \leq 8$
 - ❑ (A): $1 \leq 4$
 - ❑ (B): $5 \leq 5$



All heuristic values are admissible.

Check the Admissibility?

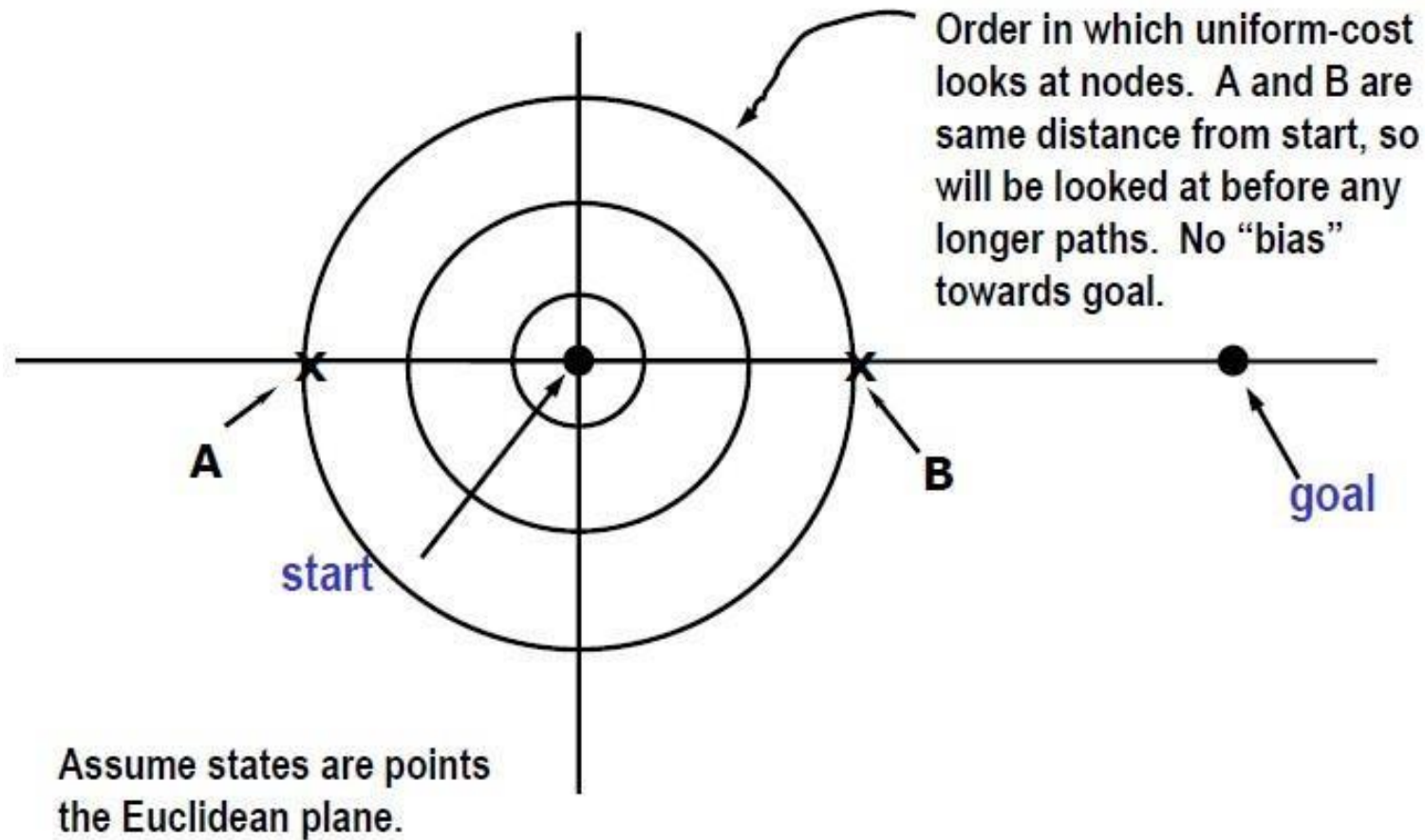
- Given the **heuristic values** and **lengths** in the Figure, are the heuristic values in the table admissible?
- **A = OK**
- **B = OK**
- **C = OK**
- **D = not OK, needs to be ≤ 2 S = not OK, should be 0.**
 - but the value of S would have no ill effect



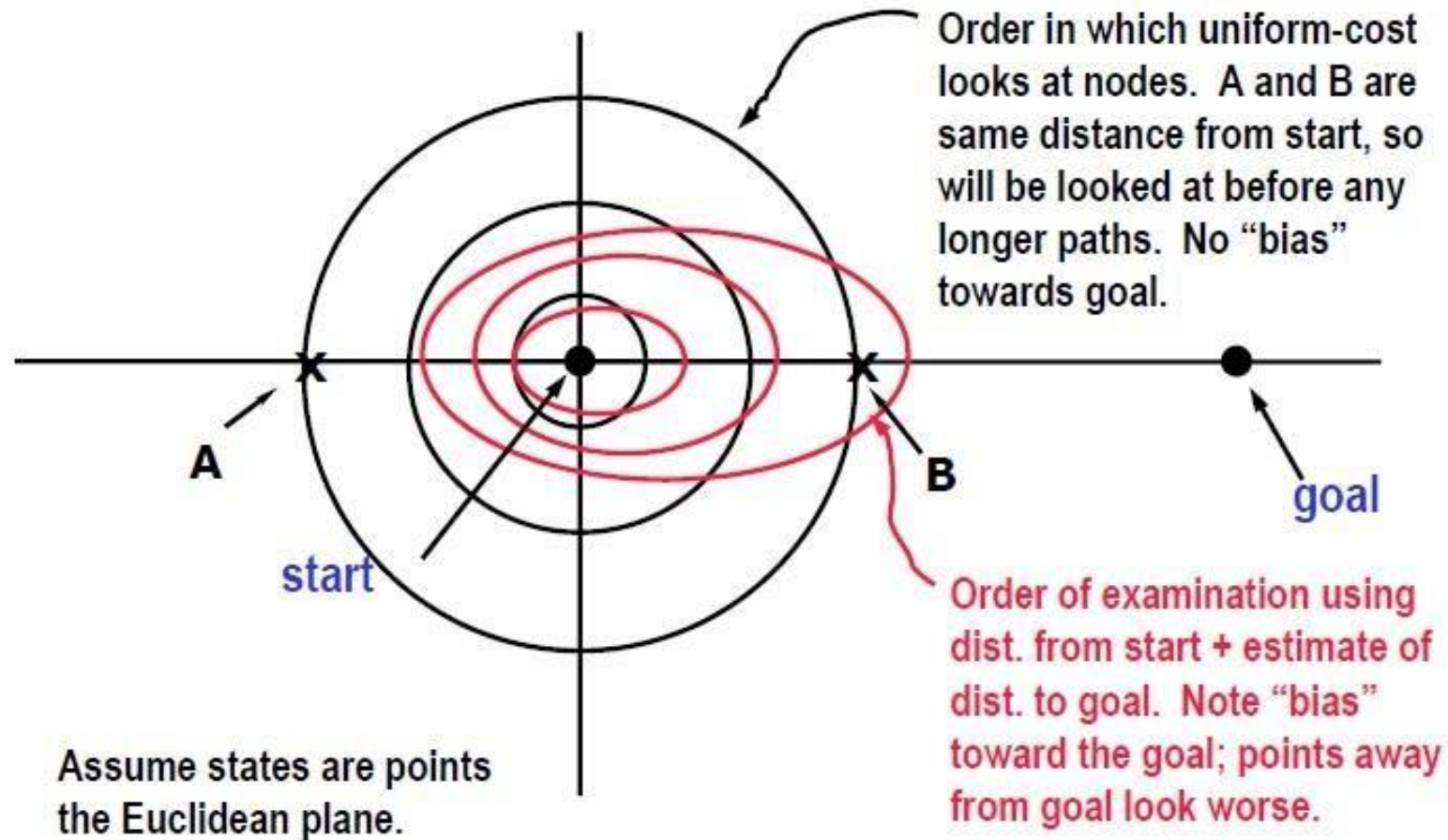
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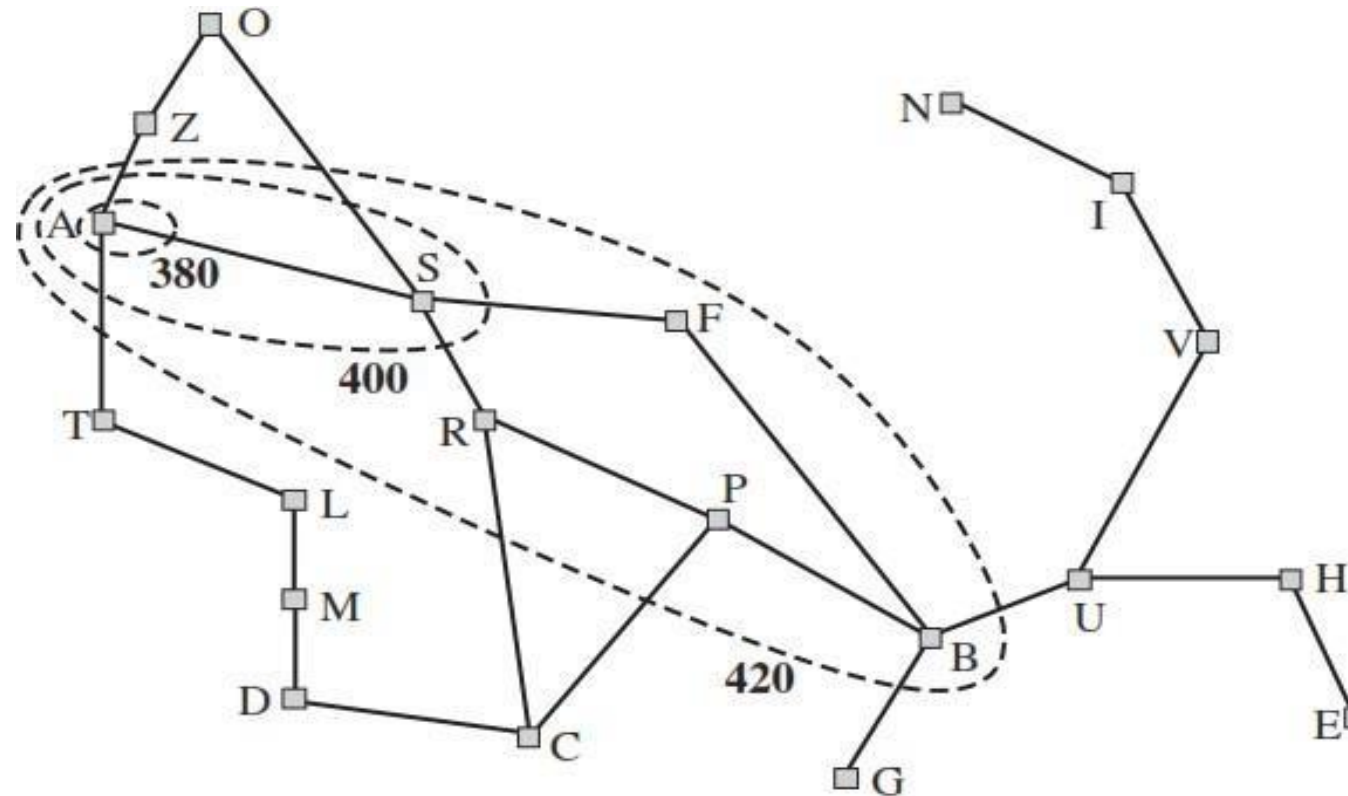
Why use estimate of goal distance?



Why use estimate of goal distance?

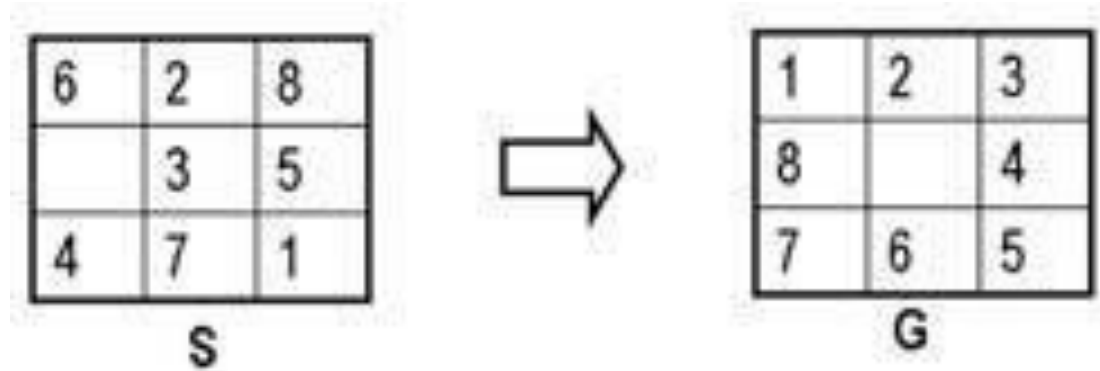


Why use estimate of goal distance?



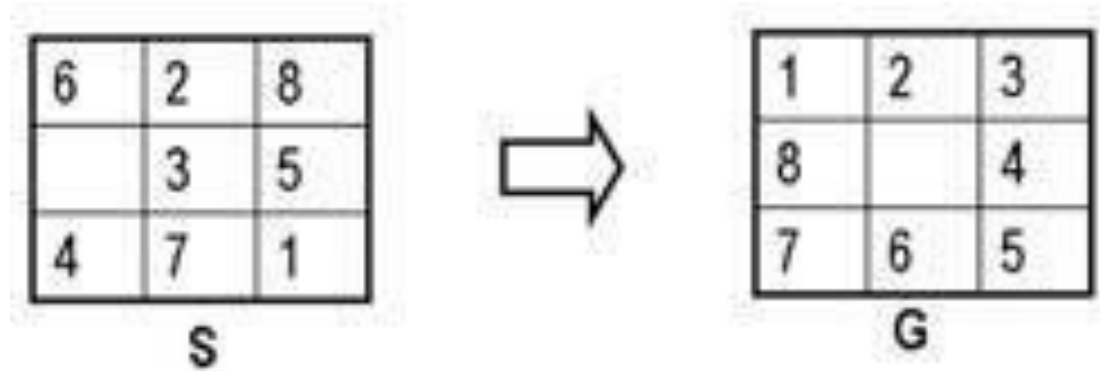
Map of Romania showing contours at $f = 380$, $f = 400$, and $f = 420$, with Arad as the start state.

Admissible Heuristics 8-Puzzle



- **The goal** is to arrange the pieces as in the goal state on the right.
- **A move** in this game is as sliding the "**empty**" space to one of its **nearest vertical or horizontal neighbours**.
- Move tiles to reach goal

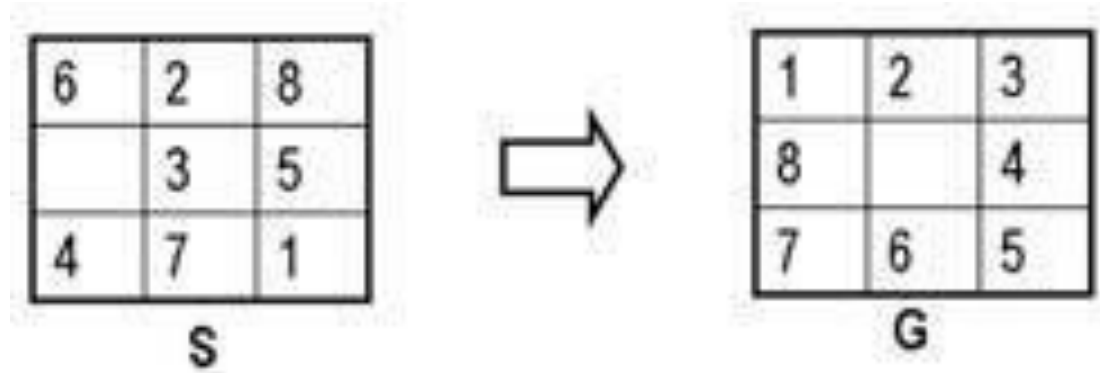
Admissible Heuristic ...8-puzzle



- Each move can at best decrease by one, the **“Manhattan distance”** of a tile from its goal.
- Manhattan distance, the metric of the Euclidean plane, is defined as

$$m((x_1, y_1)(x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$$

Admissible Heuristics



The alternative underestimates of “distance” (the number of moves) to goal:

- Number of misplaced tiles (7 in this case)
- Sum of manhattan distance, which is 17

tile 1 = 4, tile 2 = 0, tile 3 = 2, tile 4 = 3, tile 5 = 1, tile 6 = 3, tile 7 = 1, tile 8 = 3

Admissible Heuristics: Dominance

- If $h_2(n) \geq h_1(n)$ for all n (both admissible) then h_2 **dominates** h_1 and is better for search

$d = 14$ IDS = 3,473,941 nodes

$A^*(h_1) = 539$ nodes

$A^*(h_2) = 113$ nodes

$d = 24$ IDS $\approx$ 54,000,000,000 nodes

$A^*(h_1) = 39,135$ nodes

$A^*(h_2) = 1,641$ nodes

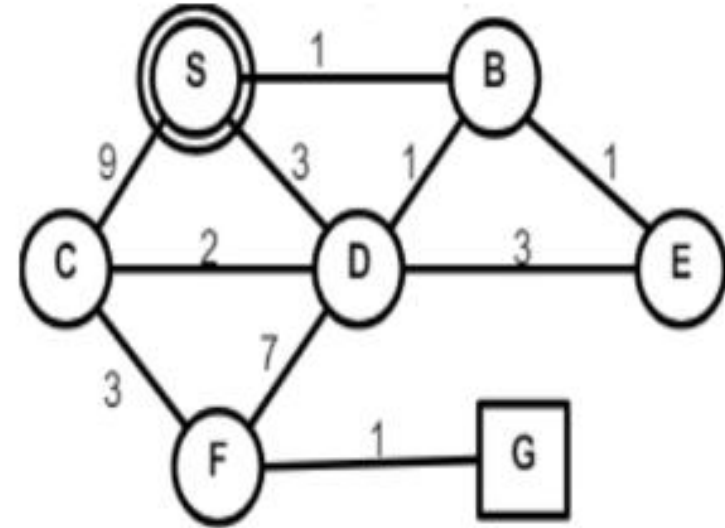
- Given admissible heuristics $h_1 \dots h_b$,

$$h(n) = \max[h_1(n) \dots h_b(n)]$$

is also admissible and dominates the other heuristics.

Dominance Example

Node	h1	h2	h3	Actual
S	5	5	8	8
B	4	5	7	7
C	3	2	4	4
D	1	4	6	6
E	3	4	8	8
F	0	1	1	1
G	0	0	0	0



Which of the following statements are true?

1. h1 dominates h2.
True/ False
2. h1 dominates h3.
True/False
3. h3 dominate h1.
True/False

Node	h1	h2	h3	Actual
S	5	5	8	8
B	4	5	7	7
C	3	2	4	4
D	1	4	6	6
E	3	4	8	8
F	0	1	1	1`
G	0	0	0	0

Which of the following statements are true?

1. h1 dominates h2.

True/ False

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True/False

3. h3 dominate h1.

True/False

Node	h1	h2	h3	Actual
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G	0	0	0	0

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G	0	0	0	0

Consistency

- To implement A* Search, **heuristic h needs to be consistent** (sometimes monotonicity):
- The goal states have a heuristic estimate of zero.

$$h(n_i) = 0 \quad \text{if node } n_i \text{ is the goal}$$

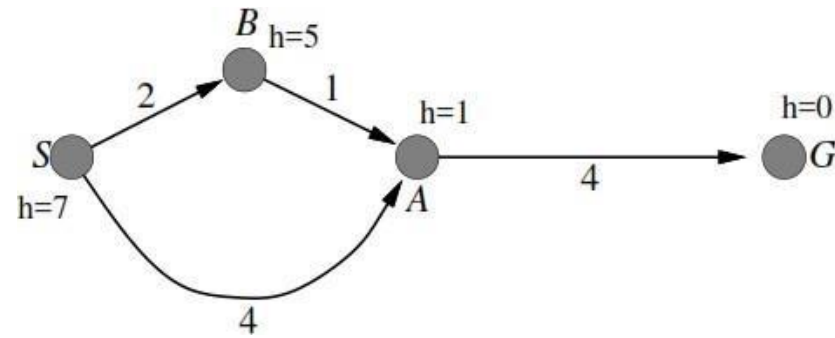
- *The difference in the heuristic estimate between one state and its descendant must be less than or equal to the actual path cost.*

$$h(n) \leq c(n, a, n') + h(n')$$

$$h(n) - h(n') \leq c(n, a, n')$$

where n' is the successor of n .

Example



- heuristic h is consistent, if

$$h(n) - h(n') \leq c(n, a, n')$$

- Consistency:

- $S \rightarrow A: 7 - 1 \leq 4$
- $S \rightarrow B: 7 - 5 \leq 2$
- $B \rightarrow A: 5 - 1 \leq 1$
- $A \rightarrow G: 1 - 0 \leq 4$

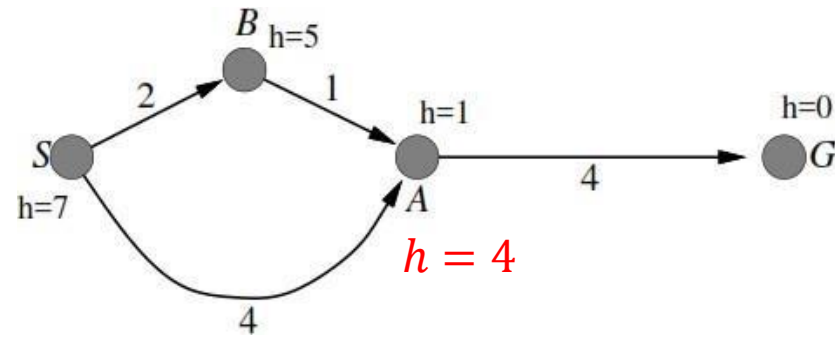
Not Consistent

Consistent

Not Consistent

Consistent

Example



$$h(n) - h(n') \leq c(n, a, n')$$

- To make them admissible and consistent, the heuristic value of **A** is changed to **4**.

- **Consistency:**

- | | |
|-----------------------------------|------------|
| • $S \rightarrow A: 7 - 4 \leq 4$ | Consistent |
| • $S \rightarrow B: 7 - 5 \leq 2$ | Consistent |
| • $B \rightarrow A: 5 - 4 \leq 1$ | Consistent |
| • $A \rightarrow G: 4 - 0 \leq 4$ | Consistent |